Smart India Hackathon 2025

Problem Statement ID : 25071

Problem Statement Title :
AI-Based Rockfall Prediction and Alert System for Open-Pit Mines

Theme : Disaster Management

PS Category : Software

Team ID : 86220

Team Name : The Rolling Stones

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AI-ML System Using DEMs, Drones & Sensors for Real-Time Rockfall Prediction and Alerts

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Problem Identification

Unpredictability — Cumulative processes, no single trigger; fractured rocks fall without warning.

Complexity — Single ML models fail; need hybrid physics + data approaches.

Monitoring & Technology Gaps — Visual checks are late & unreliable ; Lidar scans not predictive

Drivers of Rockfall Instability

Slope geometry : Steeper slopes - more shear stress

Fractured or weathered rocks have lower strength

External triggers: Rainfall, seismic activity, or freeze-thaw cycles

Mohr–Coulomb criterion helps quantify when shear strength is exceeded and failure (rockfall) will occur.

Pore Pressure — Water pressure within rock/soil pores; high values weaken slopes and raise rockfall risk. IoT piezometers provide real-time measurements.

Key trigger during rainfall and thaw cycles : used as a significant PREDICTIVE FEATURE

Our Solution and its Core Idea

- Combine **physics-based block models with ML** to predict rockfall probability, frequency, and magnitude.
- **Zone-wise scoring**: derive base risk from terrain features, updated daily with weather and sensor inputs.

Hybrid AI-ML Prediction

Multi-Source Data Integration

- **Merge** DEMs, UAV imagery, geotechnical sensors, and environmental data
- Capture **both fixed terrain factors** (slope, rock type, fracture density) **and dynamic triggers** (rainfall, pore pressure, weather).

UAV Segmentation + Image-to-DEM

- **UAV Segmentation (deep learning)** detects cracks, loose blocks, seepage zones, and vegetation stress.
- **Image-to-DEM (photogrammetry, SfM)** reconstructs slope geometry (angle, roughness, elevation).
- Together, they provide **surface + terrain insights**

Automated Alerts

- **Threshold breaches** trigger SMS, email, and app notifs
- **Early warnings** (30+ min) enable evacuation; mine holidays on high-risk days.

Interactive Dashboards

- **Real-time risk maps**, zone classifications (green/yellow/red), and trend forecasts.
- **Evacuation radii** visualized on rockfall kinematics.

Model Work Flow

Data — IoT, weather, SAC, image data normalized & transformed into statistical + temporal features with vulnerability scores

Modeling — An **ensemble classifier**, trained on historical logs involving Keras API building the NN to identify complex patterns

Risk Scoring — Model outputs a risk index (0–10), classified as **Safe | Caution | High | Critical.**

Technical Approach

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DEM Terrain geometry for long-term hazard mapping

UAV Surface instability cues like cracks / vegetation

IOT sensors Fixed and dynamic triggers

SAC signals Seismic activity cues

>>> **DEM + UAV + IOT/weather data + SAC seismic signals → predictive features that feed ML risk model → risk score**

>>> **Dashboard - live monitoring, decision support**

Image-to-DEM Pipeline

Generate DEMs from drone/satellite RGB images

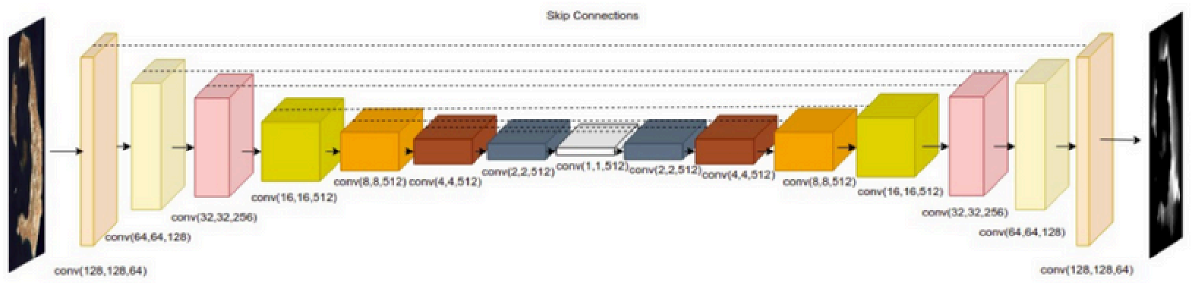
Generator (U-Net with skip connections): Encoder–decoder with skip connections preserves global structure (edges, slopes, terrain alignment).

Discriminator (PatchGAN): Evaluates DEM realism patch-by-patch for locally consistent outputs

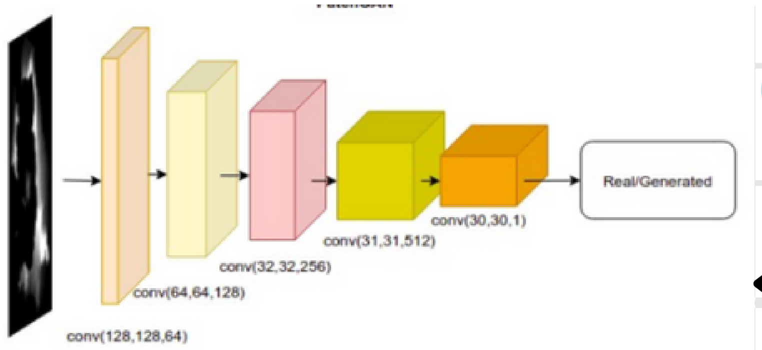
Training Process: Generator creates realistic DEMs while discriminator spots fakes, pushing the generator to improve.

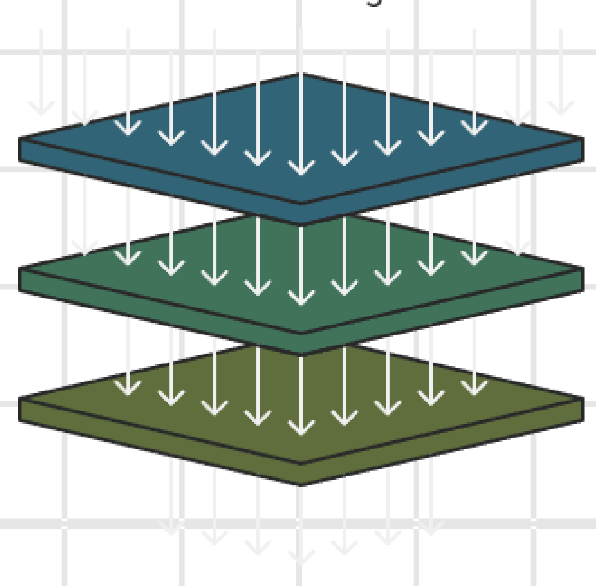
Method: Conditional Generative Adversarial Network (cGAN) (pix2pix architecture)

Generator



Discriminator





Interactive Dashboard

Functionality

1) **Real-Time Monitoring** —Ingest vibration, slope, gas, weather, battery, and personnel data.

2) **Risk Detection & Alerts** —ML-based zone scoring for automated alerts & escalation.

3) **Geospatial Visualization** — GIS maps with color-coded zones, sensor overlays, safety radii.

4) **Decision Support** — Emergency Stop, handover notifications, and KPI tiles .

Outcomes

- **Early detection, prioritized alerts**
- **Multi-site, multi-sensor support**
- **Faster response, less downtime**
- **Trends & predictive maintenance**

UAV Segmentation

CNN-based DeepLabV3+ encoder–decoder for segmentation.
Pixel-wise classification of slope hazards

Architecture

- Encoder: multi-scale feature extraction
- Decoder: segmentation masks w/ skip connections
- Speed optimizations: remove heavy atrous convs, add shallow layers, lightweight backbone

Training

- Augmentation: flips, rotations, scaling
- Loss: cross-entropy, IoU-based

Evaluation

mIoU, MPA, FPS

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Feasibility & Viability

Current Solutions

Manual Inspections: Cracks, seepage, loose rock.
Instruments: Low-frequency/manual logs.
LiDAR/Scans: High-res but costly, periodic.

Limitations

Unpredictability: Missed precursors, worker risk.
Latency: Post-event, not real-time.
Fragmented Systems: No data integration.
Cost/Scale: High cost, poor scalability.

Strategies to overcome challenges

Reliable Sensors

Redundant units, anomaly checks, edge buffering.

Model Performance

Regular retraining with fresh data, probability calibration.

Ready & Scalable

Simple UI, operator training, role access, containerized deployment.

Reduce false alerts

Apply temporal smoothing and require multi-sensor confirmation to filter noise.

Feasibility Analysis of our solution

Technical feasibility

Real-time IoT
Low-cost sensors stream live via MQTT/REST.

AI/ML Risk Scoring
Robust ensemble classifier (Random Forest, XGBoost, and a Neural Network (MLP)) built with the API Keras for rockfall prediction.

Geospatial View
Simple GIS overlays (risk zones + personnel) for intuitive decision-making.

Decision Support
Automated alerts and emergency controls reduce human dependency.

Scalable Design
Lightweight, containerized backend replicable across mines.

Economic feasibility

Lower CapEx than LiDAR
Commodity IoT sensors and open-source software reduce initial investment.

Reduced downtime
Early detection prevents costly disruptions and equipment loss.

Operational savings
Automated data collection and alerting reduce manual monitoring labor.

Cloud/edge deployment flexibility
Can scale with site size, avoiding over-provisioning.

Operational feasibility

Controlled Access
Admin-only deployment with simple authentication.

Seamless Integration
Alerts, KPIs, and handovers fit existing safety workflows.

User-Friendly Dashboard
Clear geospatial views and KPIs minimize training.

Fail-Safe Control
Emergency stop and escalation keep operators in charge.

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Impact & Benefits

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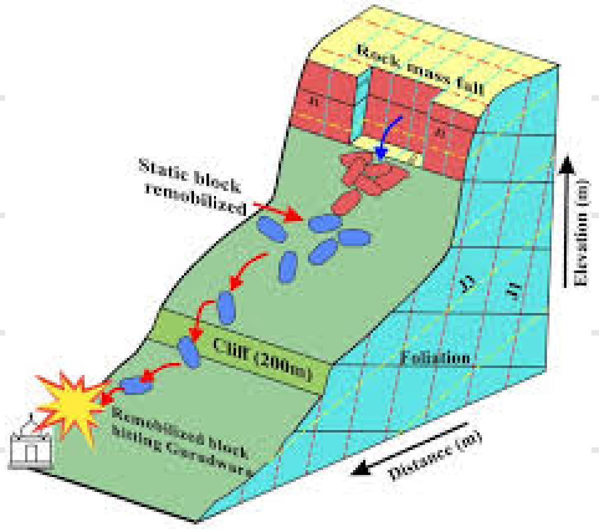
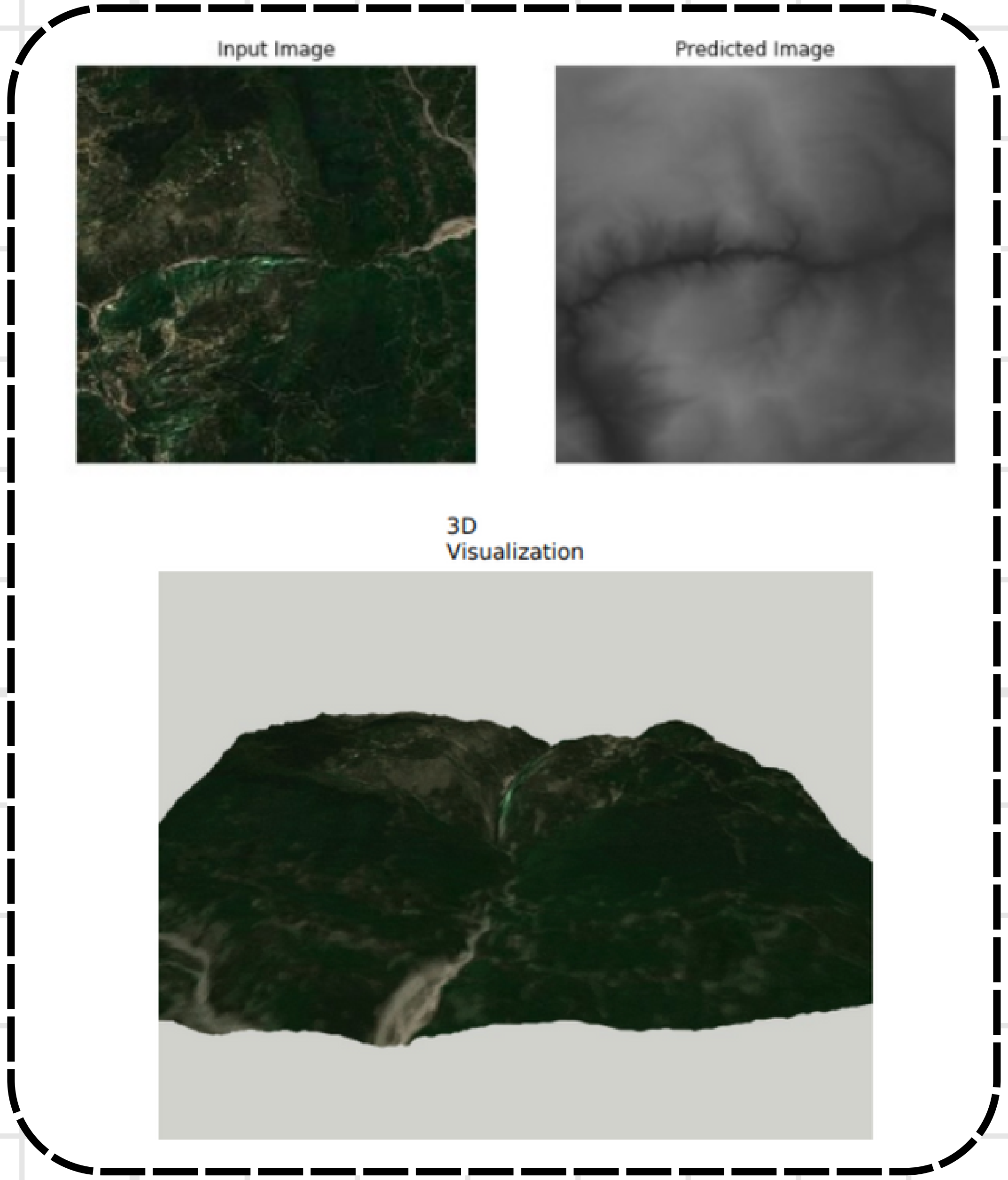


FEATURE	IMPACT	BENEFITS
Early Alerts, Evacuation Radius & Fail-Safe Controls	Reduce risk of injury/fatalities and keep operators in charge during emergencies.	Safer workforce, lower liability, stronger safety compliance.
Zone-Based Scoring & Real-Time Dashboards	Faster detection, better coordination, and prioritized hazard responses.	Reduced downtime, smoother operations, less manual effort.
Hybrid AI Models (DEM + UAV + Sensors)	Accurate detection of cracks, seepage, loose blocks with real-time updates.	Technological edge, scalable across different mines.
Low-Cost UAVs, IoT Sensors & Cloud/Edge Deployment	Affordable, real-time monitoring that scales across multiple sites.	Lower CapEx than LiDAR, cost-efficient scalability, future-proof solution.
Forecasting & Historical Risk Scoring	Enables proactive shutdowns ("safety holidays") and predictive trend analysis.	Moves mines from reactive monitoring to predictive safety



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Research & References

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Technologies evaluated, but not adopted

Synthetic Aperture Radar (SAR)

Uses satellite-borne sensors to create very detailed images of the Earth's surface. Works even at night or through clouds.

Limited vertical resolution for mine-scale terrain modeling and lower update frequency

Stereo Imagery (Sentinel/ASTER)

Uses two images of the same area from different angles to build a 3D model. Sentinel-2 (ESA/ISRO) and ASTER (NASA) provide open-access global stereo datasets.

Lacks precision for early-stage slope instability; pore pressure, rock type, and micro-movements remain unmonitored, limiting detection of fine-scale surface and subsurface changes.

Hybrid approach considered:
SAR for ground movements with Sentinel stereo imagery for 3D DEMs would enable surface motion and elevation tracking. However, limited spatial resolution and revisit frequency make it unsuitable for mine monitoring, so we chose UAV + AI-based solutions (pix2pix) instead.

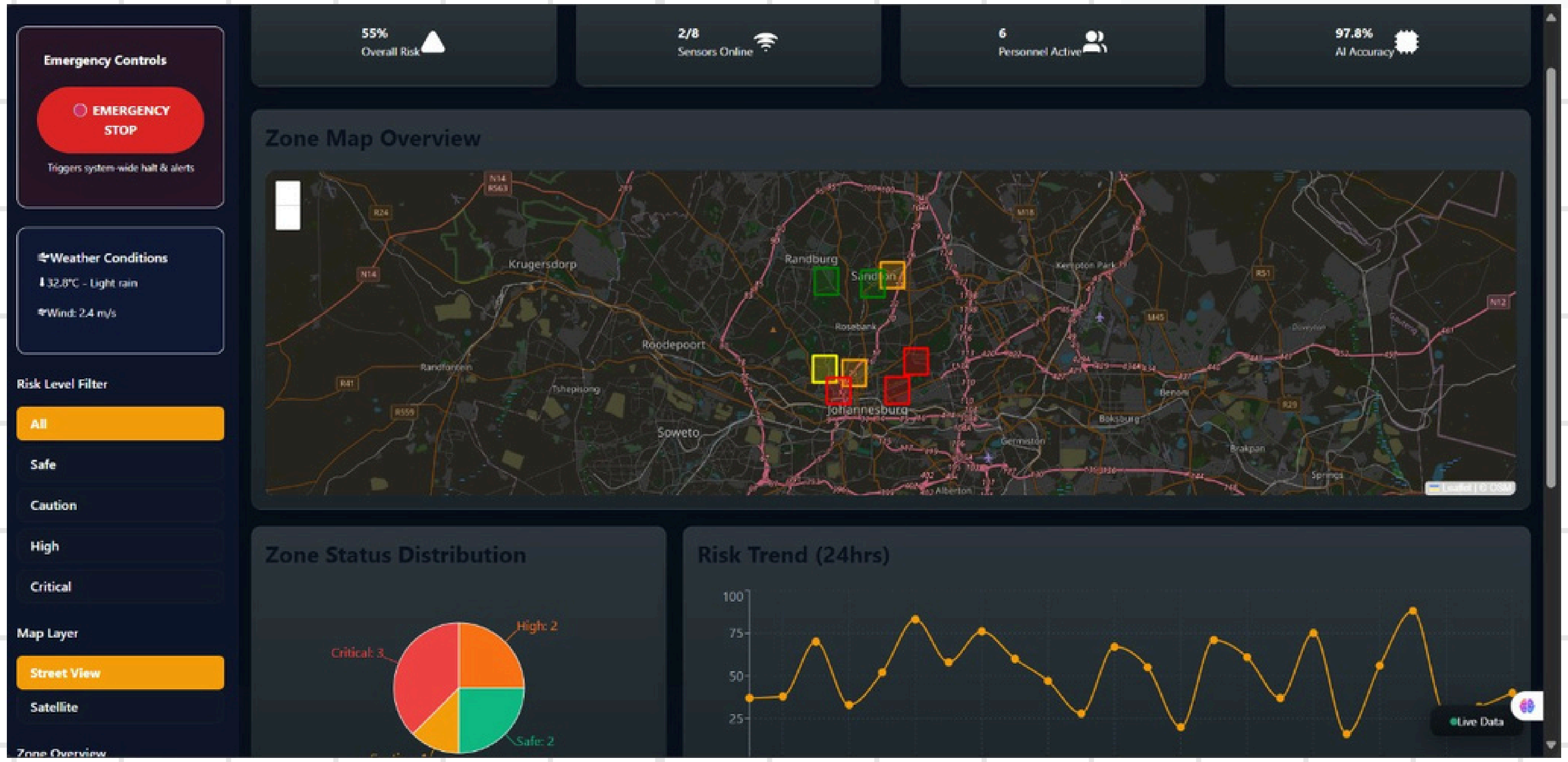
LINKS

Image-to-DEM | UAV Segmentation

Risk Model | Dashboard

Research paper | Bhukosh

भारतीय भूवैज्ञानिक सर्वेक्षण
GEOLOGICAL SURVEY OF INDIA



NLSM relevancy in rockfall prediction

Critical factors it includes :

- Slope angle & orientation
- Rock type & structural discontinuities
- Land use & vegetation cover
- Historical landslide & rockfall records

NLSM enhances both the accuracy and efficiency of any rockfall risk model by narrowing focus to geologically vulnerable zones and providing authoritative ground-truth layers.

Role in prediction model :

- Pre-classify risk zones before deploying sensors or UAVs
- Prioritize high-susceptibility areas for monitoring and data collection
- Train AI models with geologically meaningful features beyond topography
- Validate outputs from machine learning or physics-based simulations